



Topic B6

# Plant nutrition

**Concept guide · Worked application · Cambridge-style practice · Fully explained answers**

## What this guide teaches

A complete explanation of the syllabus concepts, biological mechanisms, common misconceptions, data interpretation and original exam-style practice for this topic.

## Syllabus map

|      |                      |
|------|----------------------|
| B6.1 | Photosynthesis       |
| B6.2 | Leaf structure       |
| B6.3 | Mineral requirements |
| B6.4 | Limiting factors     |

How to study: retrieve definitions, cover the application, then complete the practice paper without using the notes.

# 1. Photosynthesis

Photosynthesis transfers light energy into chemical energy by producing glucose from carbon dioxide and water; oxygen is released. Chlorophyll absorbs light. Glucose may be respired, converted to starch or cellulose, combined with nitrate to make amino acids, or transported as sucrose.

## Examiner insight

carbon dioxide + water → glucose + oxygen, in light and chlorophyll.

# 2. Leaf adaptations

A broad thin lamina gives large area and short diffusion distance. Palisade cells near the upper surface contain many chloroplasts. Spongy mesophyll air spaces increase internal exchange area. Stomata allow gas exchange; guard cells regulate opening. Xylem supplies water and phloem transports sugars.

## Examiner insight

State feature, then connect it to faster light absorption or gas exchange.

# 3. Limiting factors

Light intensity, carbon dioxide concentration and temperature can limit photosynthesis. Increasing one factor raises rate only until another becomes limiting. Temperature affects enzyme-controlled reactions; excessive heat can close stomata or denature enzymes.

## Examiner insight

A plateau does not mean photosynthesis stops; it means another factor limits the rate.

# 4. Mineral ions

Nitrate ions are needed to make amino acids and proteins; deficiency causes stunted growth. Magnesium ions are needed for chlorophyll; deficiency causes yellow leaves. Minerals are absorbed from soil by roots, sometimes by active transport.

## Examiner insight

Do not say plants obtain food from soil; they obtain mineral ions and make organic food by photosynthesis.

## Worked application

Situation: A plant has yellow leaves but adequate light and water.

Prediction: Magnesium deficiency is likely.

Reason: Magnesium is required to make chlorophyll, so deficiency reduces green pigment and photosynthesis.

# Exam technique

## High-value habits

Define terms precisely, describe visible patterns before explaining them, link structure to function, use data values as evidence, identify variables explicitly and answer the command word directly.

## Retrieval checklist

- Recall five key definitions without notes.
- Explain one process as a linked cause-and-effect chain.
- Interpret the data set and identify a limitation.
- Complete the timed practice and mark only after finishing.



# Practice Paper

Topic B6

## Plant nutrition

### Instructions

Answer all questions. Select one option for multiple choice. Use values from tables where relevant. Write explanations on the lines. Individual parts carry 1-3 marks.

Candidate name:

Class / date:

**Total marks: 50**

Suggested time: 60 minutes.

## Section A - Multiple choice

Choose the best answer for each question.

1. Which statement best defines photosynthesis? [1]
  - A the process that makes carbohydrates from carbon dioxide and water using light energy
  - B a factor in shortest supply that restricts reaction rate
  - C a pore in the leaf epidermis controlled by guard cells
  - D a green pigment that absorbs light energy
2. Which statement best defines limiting factor? [1]
  - A a pore in the leaf epidermis controlled by guard cells
  - B a factor in shortest supply that restricts reaction rate
  - C a green pigment that absorbs light energy
  - D movement of sucrose and amino acids in phloem
3. Which statement best defines stoma? [1]
  - A a green pigment that absorbs light energy
  - B movement of sucrose and amino acids in phloem
  - C a pore in the leaf epidermis controlled by guard cells
  - D the process that makes carbohydrates from carbon dioxide and water using light energy
4. Which statement best defines chlorophyll? [1]
  - A movement of sucrose and amino acids in phloem
  - B the process that makes carbohydrates from carbon dioxide and water using light energy
  - C a factor in shortest supply that restricts reaction rate
  - D a green pigment that absorbs light energy
5. Which statement best defines translocation? [1]
  - A movement of sucrose and amino acids in phloem
  - B the process that makes carbohydrates from carbon dioxide and water using light energy
  - C a factor in shortest supply that restricts reaction rate
  - D a pore in the leaf epidermis controlled by guard cells
6. Which statement about photosynthesis is correct? [1]
  - A Release of energy from glucose.
  - B Photosynthesis transfers light energy into chemical energy by producing glucose from carbon dioxide and water; oxygen is released.
  - C A factor that always stops a reaction.
  - D A chloroplast inside a palisade cell.

7. Which statement about leaf adaptations is correct? [1]
- A A factor that always stops a reaction.
  - B A chloroplast inside a palisade cell.
  - C A broad thin lamina gives large area and short diffusion distance.
  - D A mineral ion transported in phloem.
8. Which statement about limiting factors is correct? [1]
- A A chloroplast inside a palisade cell.
  - B A mineral ion transported in phloem.
  - C Movement of water only through xylem.
  - D Light intensity, carbon dioxide concentration and temperature can limit photosynthesis.
9. Which statement about mineral ions is correct? [1]
- A Nitrate ions are needed to make amino acids and proteins; deficiency causes stunted growth.
  - B A mineral ion transported in phloem.
  - C Movement of water only through xylem.
  - D Release of energy from glucose.
10. Which prediction is most appropriate when a plant has yellow leaves but adequate light and water? [1]
- A Release of energy from glucose.
  - B A chloroplast inside a palisade cell.
  - C Magnesium deficiency is likely.
  - D Movement of water only through xylem.

## Section B · Structured questions

11 Biologists use precise vocabulary when studying plant nutrition.

(a) Define photosynthesis. [2]

---

---

---

(b) Define limiting factor. [2]

---

---

---

(c) Distinguish stoma from the incorrect idea: “a chloroplast inside a palisade cell”. [2]

---

---

---

---

(d) State two important features of photosynthesis. [2]

---

---

---

---

**[Total: 8]**

12 A plant has yellow leaves but adequate light and water.

(a) State the most likely biological outcome. [2]

---

---

---

(b) Explain the outcome using a linked biological mechanism. [3]

---

---

---

---

---

(c) Predict what would happen if the relevant change were reversed. [2]

---

---

---

(d) State one assumption made in this prediction. [1]

---

---

**[Total: 8]**

13 A student records photosynthesis rate under three conditions.

| condition    | photosynthesis rate |
|--------------|---------------------|
| low light    | 4                   |
| medium light | 11                  |
| high light   | 12                  |

(a) Calculate the difference between the highest and lowest recorded values. [2]

(b) Describe the pattern in the results. [2]

---

---

---

(c) Explain the pattern using biological knowledge. [3]

---

---

---

---

(d) State one limitation of drawing a conclusion from these data. [1]

---

---

**[Total: 8]**

14 A student plans to change lamp distance to investigate photosynthesis in pondweed.

(a) Identify the independent variable. [1]

---

---

(b) Identify the dependent variable. [1]

---

---

(c) State two control variables. [2]

---

---

---

(d) Describe how to obtain reliable results across a suitable range. [3]

---

---

---

---

(e) Suggest one specific improvement. [1]

---

---

**[Total: 8]**

15 Apply your knowledge of plant nutrition to an unfamiliar context.

(a) Define chlorophyll. [2]

---

---

---

(b) Explain the biological importance of limiting factors. [3]

---

---

---

---

(c) Correct this statement: “movement of water only through xylem”. [2]

---

---

---

---

(d) State one further question that could be investigated. [1]

---

---

**[Total: 8]**



# Worked solutions

**Remove or fold back this section while completing the practice paper.**

## Marking routine

Mark each part separately. Award points for accurate biology, use of evidence and linked reasoning, then diagnose the first error.

## Solutions · Multiple choice

- 1 A. The process that makes carbohydrates from carbon dioxide and water using light energy.
- 2 B. A factor in shortest supply that restricts reaction rate.
- 3 C. A pore in the leaf epidermis controlled by guard cells.
- 4 D. A green pigment that absorbs light energy.
- 5 A. Movement of sucrose and amino acids in phloem.
- 6 B. carbon dioxide + water → glucose + oxygen, in light and chlorophyll.
- 7 C. State feature, then connect it to faster light absorption or gas exchange.
- 8 D. A plateau does not mean photosynthesis stops; it means another factor limits the rate.
- 9 A. Do not say plants obtain food from soil; they obtain mineral ions and make organic food by photosynthesis.
- 10 C. Magnesium is required to make chlorophyll, so deficiency reduces green pigment and photosynthesis.

## Solutions · Structured questions

11. (a) the process that makes carbohydrates from carbon dioxide and water using light energy [2]. (b) a factor in shortest supply that restricts reaction rate [2]. (c) a pore in the leaf epidermis controlled by guard cells; not a chloroplast inside a palisade cell [2]. (d) Credit any two accurate features from the guide: Photosynthesis transfers light energy into chemical energy by producing glucose from carbon dioxide and water; oxygen is released. Chlorophyll absorbs light. Glucose may be respired, converted to starch or cellulose, combined with nitrate to make amino acids, or transported as sucrose. [2].

12. (a) Magnesium deficiency is likely [2]. (b) Magnesium is required to make chlorophyll, so deficiency reduces green pigment and photosynthesis [3]. (c) Accept the biologically justified opposite trend when the causal factor is reversed [2]. (d) One valid assumption, such as other relevant conditions remaining constant [1].

13. (a)  $12 - 4 = 8$  [2]. (b) The values change from 4 to 12, with the highest value 12. [2]. (c) Light limits at first, then another factor causes the plateau [3]. (d) Only three conditions / no repeats / small dataset / uncontrolled variation [1].

14. (a) Light intensity or distance [1]. (b) Oxygen volume or bubbles per minute [1]. (c) Same plant length, temperature, carbon dioxide and time [2]. (d) Use several values across a sensible range, repeat each condition, identify anomalies and calculate means [3]. (e) Measure gas volume with a gas syringe and use a heat shield [1].

15. (a) a green pigment that absorbs light energy [2]. (b) Award linked points from: Light intensity, carbon dioxide concentration and temperature can limit photosynthesis. Increasing one factor raises rate only until another becomes limiting. Temperature affects enzyme-controlled reactions; excessive heat can close stomata or denature enzymes. [3]. (c) Correct statement: translocation means movement of sucrose and amino acids in phloem [2]. (d) Any testable relevant biological question [1].

### Self-assessment

Classify the first lost mark as definition, biological knowledge, causal link, data use, experimental design or communication.

Evidence used: Cambridge IGCSE Co-ordinated Sciences 0654 syllabus 2025-2027 and June 2026 papers, mark schemes and Principal Examiner Report supplied by the user. All practice questions are newly written.